ASR-Aware Unified Fusion of Three Speech Enhancers for Noise Event Removal on Vaani

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***Abstract*—We remove noise events from Indic field speech to maximise Combined = SI-SDR + 100× the word error rate reduction (ΔWER) of the mandated SraVaani-1.0 ASR. Every clip takes the same path: three public enhancers of different families (MossFormerGAN, a 104M band-split RNN, TF-Locoformer) are fine-tuned *through the frozen scoring ASR* on a test-length single-recording stream, and a 0.34M-parameter network fuses them per time-frequency cell, trained with the same losses plus a deletion-weighted minimum word error (MWER) risk. The final entry scores Combined 19.897 (SI-SDR 15.793 dB, pooled ΔWER +4.104 pp). Out of sample, it recovers 53% of the clean-speech word gain on test-length clips; 89% of the words still lost are deletions under events.**

I. Training Data and Preprocessing

Vaani validation set. The organisers' validation release provides 695 synthetic pairs (clean reference and noisy mixture, 4.5 s median) and 1,806 natural recordings with human transcripts (27 languages). A speaker hash splits the pairs 70/30: 486 train the *held-out-protocol* members and 209 are never seen by them; the shipped MossFormerGAN and BSRNN are retrained on all 695. Natural clips are never trained on; a fixed 30% hash subset (555 clips, 7,834 words) measures natural ΔWER.

Noise events (augmentation only). Noisy minus clean, cut at the annotated spans, gives 2,213 events (23 min, 7 categories), split by the speaker hash; they only synthesise training mixtures.

Augmentation pipeline. Each crop (2 s for MossFormerGAN, 4 s otherwise) is, with probability 0.5, a real pair, else a clean reference re-mixed with 1–3 random events at an SNR drawn from the real pairs (normalised if above full scale). A *long stream* adds one micro-batch per step of 12–18 s windows of single-speaker *beds* (clean recordings), with events pasted as in the organisers' mixtures (Poisson 0.5 events/s, log-normal gain, median 0.93 on the event's amplitude, σ=0.8) at reference levels; it trains only the ASR losses (rationale 2 in Section II).

External data (Table I). Two public corpora supply clean speech *beds*: Vaani-transcription-part (the long beds and 908 fusion beds) and IndicVoices-R (1,592 fusion beds). Competition clips and every recording behind a validation or Phase 1 test clip were excluded by name; beds were screened with Silero-VAD [3] and TorchAudio-SQUIM [4]. Each clean bed's greedy SraVaani transcript is its target (self-distillation). No test audio is used; no test clip shares 30% of its word 8-grams with any of the 6,450 training recordings.

TABLE I
Every dataset we used, and its role in the final solution. No test audio, external noise or synthetic (TTS) speech is used anywhere.

Data (source, version)	Used	How	Final
Organiser data (Vaani validation release)			
Synthetic pairs	695 pairs, 1.0 h	member training; 209 held out	yes
Noise events (noisy − clean)	2,213, 0.4 h	training mixtures only	yes
Natural recordings	555 of 1,806	evaluation only	no
External data, gated, used with all competition recordings excluded			
Vaani-transcription-part [1] (`d2acadf`)	1,450 long beds, 5.7 h	long ASR stream, all members	yes
	908 beds, 3.2 h	fusion training mixtures	yes
IndicVoices-R [2] (`5f4495c`)	1,592 beds, 6.7 h	fusion training mixtures	yes
	198 h more	member training (Table II)	no
Simulated from the above (events pasted onto clean beds)			
fresh2	2,500 clips, 9.9 h	fusion training	yes
valset	1,200 clips, 4.2 h	member screening; fusion training	yes
valset_long	360 clips, 1.3 h	evaluation only	no

Simulated sets: *fresh2* is 8–18 s single recordings (0.65 events/s) on beds no member saw; *valset* glues held-out speakers' references (33.8k words); *valset_long* has one recording per unseen speaker.

II. Proposed Method

Design rationale. Four early measurements shaped the system. (1) *Waveform quality does not buy words*: on valset, at the same SI-SDR (15.8 dB), clean speech with the events turned down 20 dB recovers 92% of the clean-speech word gain, our first blend only 32%, and every architecture we trained kept only 47–57% of the speech energy in cells where an event dominates. So we train through the scoring ASR. (2) *The test is longer, and about as hard*: test clips match our pairs in estimated SNR (-3.4 vs -2.5 dB median) but last 13 s against 4.5 s (lengths and estimated SNRs of the unlabelled test inputs are the only test statistics we used), and SraVaani's encoder states on 2 s crops differ from their full-context values (cosine distance 0.24; 0.08 at 8 s). So the ASR losses also see 12–18 s single recordings. (3) *Diversity comes from families, not seeds*: errors of two seeds of one model correlate 0.95–0.99, a new family's errors only 0.69 with our mix, and on the board each of two families alone scored 0.6–1.0 Combined below their blend. So we fuse three families per cell. (4) *Lost words are deletions* (71% of the gap to clean speech early on, 89% in the final system), so the word-error risk charges deletions double.

Contributions. Beyond mimic losses [5], observation adding [6], [7] and sampled MWER [8], [9]: (i) enhancer losses on the scoring ASR's own TDT lattice, with a deletion-weighted MWER via REINFORCE [10]; (ii) a test-length stream for them; (iii) a per-cell fusion with an anchored observation term.

TABLE II

WHAT WE TRIED. EACH CHANGE IS MEASURED PAIRED AGAINST ITS OWN CONTROL ON A SET THE COMPARED MODELS NEVER TRAINED ON (SIM = VALSET, 33.8K WORDS; LONG = VALSET_LONG; H/O = 209 HELD-OUT PAIRS, FUSIONS 2-FOLD OUT-OF-FOLD; NAT = 555 NATURAL CLIPS). WORDS: ΔWER CHANGE IN PP [95% CI]; ° CI INCLUDES ZERO. BOARD: CHANGE IN COMBINED OF A SINGLE UPLOAD (NOISE ABOUT ±0.1). MFGAN = MOSSFORMERGAN, TF-LOCO = TF-LOCOFORMER. †EXCEPTIONS TO THE KEEP RULE: THREE MEMBERS KEPT FOR SIMPLICITY AT EQUAL WORDS; THE MWER FINE-TUNE KEPT ONLY AS ONE OF THREE SOUP INGREDIENTS.

Change	Model	Set	Words (pp)	SI-SDR (dB)	Board	Kept
ASR-aware losses (rationale 1, 2, 4)						
Encoder distance, weight 30	MFGAN	sim	+0.30 [0.06, 0.53]	0.00		yes
Encoder distance, weight 30, 8k steps	MP-SENet	sim	+0.54 [0.24, 0.81]	0.00	+0.18	yes
+ transcript NLL, weight 2.5	BSRNN 38M; 104M; MFGAN	sim	+0.61 [0.36, 0.85]; +0.43 [0.22, 0.65]; +0.24 [0.03, 0.50]	+0.05; 0.00; 0.00		yes
Heavier weights (encoder 100; NLL 7.5)	MP-SENet; BSRNN 38M	sim	+0.05; −0.09°	−0.11; −0.07		no
Long single-recording stream (2 seeds)	BSRNN 38M	sim; long	+1.47 [1.18, 1.78]; +0.38 [0.14, 0.62]	+0.30; –		yes
Long stream also on MFGAN and TF-Loco	3 members	long	+0.33 [0.07, 0.60]	+0.23	+0.11	yes
MWER (weight 3) on a member	TF-Loco	long	−0.17°	–	+0.02	soup†
MWER in the fusion	fusion	long	+0.26 [0.10, 0.43]	−0.02	+0.04	yes
Words-first fusion (ASR losses ×3)	fusion	nat; long	+0.14 [0.01, 0.26]; +0.05°	−0.19 (long)	−0.02	no
Learned per-cell fusion vs fixed mix	fusion	h/o	+0.04°	+0.058 [0.021, 0.095]	+0.13	yes
Members and data (rationale 3)						
Add TF-Locoformer (error corr. 0.69 vs 0.82–0.90)	mix	long	–	+0.79 (351/360 clips)	+1.37	yes
Three members instead of eight	fusion	long	−0.04°	−0.11	+0.02	yes†
Drop MossFormerGAN (two members)	fusion	board			−0.08	no
Weight soup of three fine-tunes	TF-Loco	h/o	+0.17°	+0.085 [0.056, 0.118]	+0.03	yes
198 h more clean speech (IndicVoices-R)	4 families	h/o	+0.10°	−0.05 to −0.17		no
Other pretrained models (Table III)	mix	h/o	−3.5 (UniPASE)	−2.4 to −8.6		no
Targets, fusion and post-processing						
Speech-intact target (noise left at −20 dB)	TF-Loco	long	−0.20°	−4.5		no
Two-stage member; ASR-feature inputs; routing probe	fusion	long	−0.02 to +0.13°	−0.06 to +0.02		no
Louder output (peak normalisation)	post	sim	+0.71 [0.47, 0.97]	0.00	−0.19	no

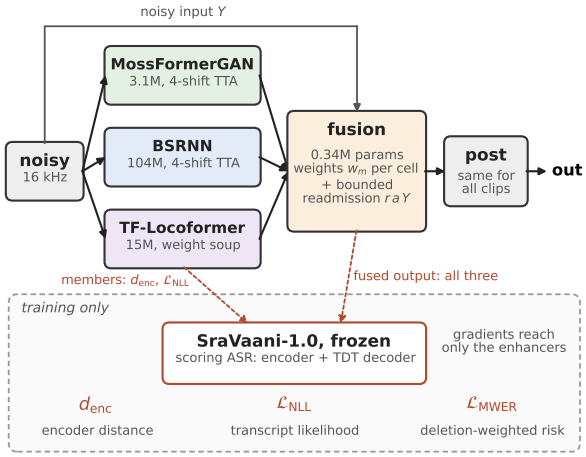


Fig. 1. Every clip takes the same path (top). In training only, the frozen scoring ASR scores the members and the fused output (bottom); MWER is used in the fusion and one TF-Locoformer fine-tune.

Members (Fig. 1, Table III) are fine-tuned with

$$\mathcal{L} = -\text{SI-SDR}(\hat{s}, s) + 30\, d_{\text{enc}}(\hat{s}, s) + 2.5\, \mathcal{L}_{\text{NLL}} + \lambda\, \mathcal{L}_{\text{MWER}}, \quad (1)$$

plus 15× the compressed-spectrum L1 loss for MossFormerGAN. d_{enc} is the mean per-frame cosine distance between SraVaani's final encoder states of the estimate and of clean speech (differentiable front end). $\mathcal{L}_{\text{NLL}}$ is the TDT [11] negative log-likelihood, on the estimate's lattice, of the teacher tokens inside the crop (decoded once on the full clean recording, with their frame times), divided by the token count; paths may start and end within 240 ms of a crop edge. $\mathcal{L}_{\text{MWER}}$ draws $N{=}4$ ancestral samples per window of 8 s or more (the long stream) and weights each sample's log-likelihood by its word cost against the teacher (deletions cost 2) minus the mean of the four, divided by $N\times$ reference words. $\lambda{=}3$ in one of the three TF-Locoformer fine-tunes and in the fusion, otherwise 0. The ASR is frozen; gradients reach only the enhancer. *Training* (AdamW, warm-up, cosine decay, clipping at 5, weight EMA 0.999–0.9995): MossFormerGAN (2 s crops, batch 4) for 40k steps with SI-SDR and L1 on the 486 pairs, then on all pairs 12k +Enc, 4k +NLL, 4k +long stream (LR 5e-5); BSRNN (4 s, batch 8) for 30k +Enc (LR 1e-4), 6k +NLL and 6k +long stream; TF-Locoformer (4 s, batch 4, LR 1e-4) for 20k steps with SI-SDR, Enc, NLL and the long stream from MERL's checkpoint on the 486 pairs, then three 4k fine-tunes (with MWER, plain, events 6 dB louder) averaged into one model, a weight soup [12].

TABLE III

PRETRAINED MODELS IN THE FINAL SYSTEM AND THEIR ROLES.

Model (source, version)	Params	Role
MossFormerGAN_SE_16K (ClearVoice [13] 0.1.2)	3.1M	member 1, 4-shift TTA
BSRNN large-double, VCTK+DNS+WHAMR! [14]	104M	member 2, 4-shift TTA
`wyz/vctk_dns2020_whamr_bsrnn_large_double_noncausal, 91epoch.pth`		
TF-Locoformer M, DNS-2020 (MERL [15])	15M	member 3, weight soup
`github.com/merlresearch/tf-locoformer, egs2/dns_ins20, valid.loss.ave_5best.pth`		
SraVaani-1.0 [16] (rev. `f5dd535`)	–	frozen loss, transcripts
Silero-VAD 6.2.3 [3], SQUIM (torchaudio 2.9.0) [4]	–	bed screening

Tried, not in the final system (Table II): MP-SENet, FRCRN, TF-GridNet, CleanUNet, UniPASE, Sidon, ZipEnhancer, SEMamba, MossFormer2-48k, USES, Conv-TasNet.

Fusion. Member outputs are least-squares scaled onto the input (512-point Hann STFT, hop 128). From the log powers of the members, the input and a fixed mix $B = \sum_m w_m^0 X_m$, each member's phase and level relative to B and the frequency position, eight residual blocks of dilated 3×3 convolutions (64 channels) predict per-cell softmax weights w_m (initialised at $w^0 = (0.2, 0.35, 0.45)$) and a complex readmission r of the noisy cell ($|\Re r|, |\Im r| \le 0.2$), scaled by how far the mix moved that cell:

$$\hat{S} = \textstyle\sum_m w_m X_m + r\, a\, Y, \quad a = \min\big(1, |Y - B|/|Y|\big). \quad (2)$$

It trains for 3,000 steps (LR 5e-4) on 8 s crops of out-of-sample member outputs (*twins*: the same recipes trained on the 486 pairs only; fresh2, the 209 pairs and valset), through the shipped post-processing. At inference it combines the all-

TABLE IV

PHASE 2 BOARD HISTORY (CODABENCH UPLOAD NAMES); Δ IS AGAINST THE PARENT UPLOAD (THE ROW ABOVE UNLESS NAMED), FROM UNROUNDED SCORES. UPLOADS ABOVE THE RULE PREDATE THE UNIFIED-PIPELINE RULE AND ROUTED NATURAL CLIPS THROUGH A LIGHTER PATH. ‡TWO CHANGES IN ONE UPLOAD. OUR FIRST PHASE 2 ENTRY (ROUTED MP-SENET/FRCRN) SCORED 16.15.

Upload	Change (parent)	Comb.	Δ
p2_02	blend of MP-SENet, FRCRN, MossFormerGAN	16.33	+0.18
p2_08	ASR-encoder loss on MP-SENet (p2_02)	16.51	+0.18
p2_09	+ BSRNN family, members on all pairs‡ (p2_02)	16.97	+0.63
p2_17a	re-weighted members, shift TTA‡	17.29	+0.32
p2_18	104M BSRNN, encoder + transcript loss‡	17.48	+0.19
p2_19	learned per-cell fusion	17.60	+0.13
p2_23	fusion on members with ASR losses (p2_22, 17.55)	17.79	+0.24
p2_28	+ TF-Locoformer with ASR losses (p2_23)	19.16	+1.37
p2_32	unified pipeline, long stream on BSRNN‡	19.30	+0.14
p2_33	8-member fusion, readmission, test-length data‡	19.68	+0.38
p2_35	three members only	19.70	+0.02
p2_37	long stream on all three members	19.81	+0.11
p2_39	MWER on one TF-Locoformer	19.83	+0.02
p2_43	TF-Locoformer weight soup	19.86	+0.03
p2_44	**MWER in the fusion (final)**	**19.90**	+0.04

TABLE V

WHERE WORDS ARE LOST ON VALSET_LONG (ΔWER IN PP, 95% CI). *FUSION TRAINED ON FRESH2 ONLY, WITH THE HELD-OUT-PROTOCOL MEMBERS, SO THAT THE SET IS OUT OF SAMPLE.

valset_long (360 clips)	SI-SDR	ΔWER (share of ceiling)
Clean speech (ceiling)	–	+7.59 (100%)
Speech intact, events at −20 dB	14.39	+6.54 [5.66, 7.46] (86%)
Final recipe*	16.05	+4.06 [3.46, 4.76] (53%)
coverage 1–21/21–32/32–72%: +1.54/+4.00/+6.65 (52/56/52%)		

pair MossFormerGAN and BSRNN, which it never saw in training; this twin swap was not measured separately.

Inference. MossFormerGAN and BSRNN average four input delays (0, 41, 83, 124 samples [17]); long clips are decoded in overlapping windows (TF-Locoformer 8 s/6 s, gain-matched; BSRNN 20 s/15 s). Post-processing, identical for every clip, uses the model output where it changed the audio (smoothed residual above 1.5% of its maximum, dilated 150 ms) and keeps the input elsewhere (10 ms cross-fades), then gates at −50 dB and limits peaks to 0.99; the fusion trains through it, and shipped audio differs from the pure fusion output by −41 dB (median). There is no clip classifier, branch, whole-clip pass-through, or template, dictionary or correlation matching.

III. RESULTS AND CONCLUSION

Metric and protocol. Combined = SI-SDR [18] (synthetic clips) $+ 100 \cdot \Delta$WER (noisy minus enhanced WER, pooled over 3,178 clips). Local comparisons are paired, on sets the compared models never trained on, with a clip bootstrap (2,000 draws, 95% CI; valset_long has one clip per speaker).

Validity. The final entry is the best of 45 Phase 2 submission files, so 19.90 is optimistic; one upload resolves about ±0.1 Combined, so board steps under 0.1 rest on local intervals. Local gains did not always transfer (one word gain reversed: louder output), and seeds of a 6k-step continuation differ by ±0.35 pp. So we changed one thing per upload where possible and kept a component only if a local interval excluded zero, re-checking gains near 0.3 pp on a second seed or set; Table II marks the two exceptions.

Key findings. (i) The ASR losses raised words on every family we tried at no SI-SDR cost, and long single-recording windows gave the largest member-level gain (Table II; valset glues several speakers per clip, which is why it shows more than valset_long). (ii) The largest system gain came from a new family, TF-Locoformer, strong and least correlated with the mix; eight members or 198 h more speech did not help. (iii) On test-length clips (Table V) the final recipe recovers 53% of the clean-speech word gain at every event coverage, and 89% of the words still lost are deletions; intact speech under attenuated events keeps 86% at a *lower* SI-SDR, so the limit is per-cell gain accuracy under events, not residual noise. (iv) In natural clips word error is 25.4% in clear speech, 26.8% under local events and 32.7% under clip-long ambient noise, and ours leaves each within 0.4 pp; all noise-attributable errors amount to about 1 pp. Locally our natural gain is zero (-0.05 pp [-0.38, $+0.27$]) against $+0.54$ pp on the board; the test's natural clips may differ.

Code (training, data preparation, inference, a stage-by-stage reproduction guide) **and checkpoints** (final and every stage): https://github.com/mcjoshi/asr-aware-noise-event-removal, https://huggingface.co/manishjoshiai/asr-aware-noise-event-removal.

Limitations and conclusion. Shift averaging slows inference; most results use one seed and the scoring ASR only. Three public enhancers trained against the scoring ASR's objectives and fused per cell reach Combined 19.90 (fourth on the board), processing every clip identically.

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